

Lecture Notes on NAND Flash Memory Technology

Chapter 1: Fundamentals of NAND Flash Memory

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1 1.1 Introduction to Non-Volatile Memory

• Evolution of Memory Technologies

Memory hierarchy in modern computing systems:

- **Volatile:** SRAM (cache), DRAM (main memory)
- **Non-volatile:** Flash (storage), Emerging NVM

Key metrics comparison:

$$\text{FOM} = \frac{\text{Density} \times \text{Speed}}{\text{Power} \times \text{Cost}} \quad (1)$$

• NAND vs NOR Flash

Definition 1.1 (NAND Flash). Series-connected memory cells with:

- High density: $\sim 10F^2$ cell size
- Page-based operations
- Sequential access optimized
- Primary use: Mass storage

2 1.2 Floating Gate Transistor Physics

• Device Structure

The floating gate (FG) transistor consists of:

- Control Gate (CG): External electrode
- Floating Gate (FG): Charge storage node
- Tunnel oxide: $\sim 8 - 10nmSiO_2$
- Inter-poly dielectric (IPD): ONO stack

- **Threshold Voltage Modulation**

Threshold voltage with stored charge:

$$V_{th} = V_{th}^0 + \frac{Q_{FG}}{\alpha C_{ox}} \quad (2)$$

where:

- V_{th}^0 = intrinsic threshold voltage
- Q_{FG} = charge on floating gate
- α = coupling ratio
- C_{ox} = gate oxide capacitance

- **Coupling Ratio**

$$\alpha = \frac{C_{IPD}}{C_{IPD} + C_{tun} + C_{dep}} \quad (3)$$

Typical values: $\alpha \approx 0.6 - 0.7$

3 1.3 Charge Storage Mechanisms

- **Fowler-Nordheim Tunneling**

Theorem 3.1 (F-N Tunneling Current). The tunneling current density through a triangular barrier:

$$J_{FN} = AE_{ox}^2 \exp\left(-\frac{B}{E_{ox}}\right) \quad (4)$$

where:

$$A = \frac{q^3}{8\pi\hbar\phi_B} \approx 1.54 \times 10^{-6} \text{ A/V}^2 \quad (5)$$

$$B = \frac{8\pi\sqrt{2m^*}}{3qh}\phi_B^{3/2} \approx 2.63 \times 10^8 \text{ V/cm} \quad (6)$$

- **Channel Hot Electron Injection (CHE)**

Energy requirement for hot electron injection:

$$E_{kinetic} > \phi_B \approx 3.2 \text{ eV} \quad (7)$$

Injection efficiency:

$$\eta_{inj} = \frac{I_g}{I_d} \approx 10^{-6} \text{ to } 10^{-4} \quad (8)$$

4 1.4 Band Diagrams and Energy Levels

- **Equilibrium Band Structure**

Floating gate potential (no stored charge):

$$V_{FG} = \alpha V_{CG} + (1 - \alpha)V_{channel} \quad (9)$$

- **Programming State**

Electric field in tunnel oxide:

$$E_{tun} = \frac{V_{CG} - V_{FG} - \phi_{ms}}{t_{ox}} \quad (10)$$

For F-N tunneling: $E_{tun} > 10$ MV/cm

- **Charge Loss Mechanisms**

1. **Thermionic Emission**

$$J_{th} = A^* T^2 \exp\left(-\frac{q\phi_B}{kT}\right) \quad (11)$$

2. **Trap-Assisted Tunneling**

$$J_{TAT} = qN_t\sigma_n v_{th} n \exp\left(-\frac{E_t}{kT}\right) \quad (12)$$

5 1.5 Cell Current Characteristics

- **Subthreshold Region**

Cell current in weak inversion:

$$I_{cell} = I_0 \exp\left(\frac{q(V_{gs} - V_{th})}{nkT}\right) \left[1 - \exp\left(-\frac{qV_{ds}}{kT}\right)\right] \quad (13)$$

Subthreshold slope:

$$SS = \frac{dV_{gs}}{d(\log_{10} I_d)} = \frac{kT \ln(10)}{q} \cdot n \approx 60n \text{ mV/decade} \quad (14)$$

- **On-State Current**

Saturation current (programmed cell):

$$I_{on} = \frac{\mu C_{ox}}{2} \frac{W}{L} (V_{gs} - V_{th})^2 \quad (15)$$

Read window:

$$\Delta I = I_{erased} - I_{programmed} \quad (16)$$

6 1.6 Memory Cell States

- **Binary Storage (SLC)**

Two states:

- Erased: $V_{th} < V_{ref}$ (logical '1')
- Programmed: $V_{th} > V_{ref}$ (logical '0')

- **Multi-Level Cell (MLC/TLC/QLC)**

Definition 6.1 (Bits per Cell). For a cell storing n bits:

$$\text{Number of states} = 2^n \quad (17)$$

The required threshold-voltage window per state is

$$\Delta V_{\text{th,state}} = \frac{V_{\text{th,max}} - V_{\text{th,min}}}{2^n - 1} - \text{margin} \quad (18)$$

7 1.7 Comparison with Other NVM Technologies

- **Figure of Merit**

$$\text{FOM}_{\text{NVM}} = \frac{\text{Retention} \times \text{Endurance} \times \text{Density}}{\text{Write Energy} \times \text{Access Time}} \quad (19)$$

- **Technology Comparison**

- **NAND Flash:** Best density, block erase
- **NOR Flash:** Random access, slower write
- **EEPROM:** Byte-erasable, low density
- **Emerging NVM:** PCM, RRAM, MRAM advantages

8 Examples

Example 8.1 (Threshold Voltage Shift). Calculate V_{th} shift for 10^4 electrons stored on FG.

Given: $\alpha = 0.65$, $C_{\text{ox}} = 1.5 \times 10^{-7} \text{ F/cm}^2$

Solution:

$$Q_{\text{FG}} = n \cdot q = 10^4 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-15} \text{ C} \quad (20)$$

$$\Delta V_{\text{th}} = \frac{Q_{\text{FG}}}{\alpha C_{\text{ox}}} \quad (21)$$

$$= \frac{1.6 \times 10^{-15}}{0.65 \times 1.5 \times 10^{-7}} \quad (22)$$

$$= 1.64 \text{ V} \quad (23)$$

Example 8.2 (F-N Tunneling Time). Estimate programming time for $\Delta V_{\text{th}} = 3 \text{ V}$.

Given: $E_{\text{ox}} = 12 \text{ MV/cm}$, cell area = $0.01 \mu\text{m}^2$

Solution:

$$J_{\text{FN}} = 1.54 \times 10^{-6} \times (12 \times 10^6)^2 \exp\left(-\frac{2.63 \times 10^8}{12 \times 10^6}\right) \quad (24)$$

$$\approx 2.2 \times 10^{-3} \text{ A/cm}^2 \quad (25)$$

Charge required:

$$Q = \alpha C_{\text{ox}} A \cdot \Delta V_{\text{th}} \quad (26)$$

$$= 0.65 \times 1.5 \times 10^{-7} \times 10^{-10} \times 3 \quad (27)$$

$$= 2.93 \times 10^{-17} \text{ C} \quad (28)$$

Programming time:

$$t_{\text{prog}} = \frac{Q}{J_{\text{FN}} \cdot A} \quad (29)$$

$$= \frac{2.93 \times 10^{-17}}{2.2 \times 10^{-3} \times 10^{-10}} \quad (30)$$

$$\approx 133 \text{ } \mu\text{s} \quad (31)$$

9 Homework Problems

Assignment 1.1: NAND Flash Fundamentals

1. Coupling Ratio Analysis:

- IPD thickness: 15 nm (ONO), $\epsilon_r = 6$
- Tunnel oxide: 8nmSiO_2
- Calculate coupling ratio α
- Find sensitivity to IPD thickness variation

2. F-N Tunneling Current:

- Plot J_{FN} vs E_{ox} for 8-15 MV/cm
- Calculate charge injection rate
- Determine programming pulse width for $\Delta V_{\text{th}} = 4 \text{ V}$
- Analyze temperature dependence

3. Multi-Level Cell Design:

- Design TLC (3 bits/cell) distributions
- Total V_{th} window: 6 V
- Include 0.5 V margins between states
- Calculate read reference voltages

4. Charge Loss Analysis:

- Initial $Q_{\text{FG}} = 500$ electrons
- Trap density: $N_t = 10^{17} \text{ cm}^{-3}$
- Calculate retention time at 25°C and 85°C
- Estimate activation energy

5. Subthreshold Characteristics:

- $n = 1.3$, $V_{DD} = 3.3$ V
- Design for $I_{on}/I_{off} = 10^6$
- Calculate required V_{th} values
- Determine sensing margin

6. Energy Band Diagram:

- Draw band diagrams for program/erase/read
- Include all voltage drops
- Calculate barrier heights
- Show electron/hole paths

7. Cell Size Scaling:

- Compare 2D scaling limits
- Calculate minimum feature size F
- Analyze short channel effects
- Propose 3D solutions

8. Disturb Mechanisms:

- Program disturb: unselected cells on same WL
- Pass disturb: cells on same string
- Calculate disturb rates
- Design mitigation strategies

Next Chapter Preview: Chapter 2 will cover "NAND Flash Architecture" including array organization, peripheral circuits, and layout optimization.